Mechanics Project Assignment

Due Friday Oct. 16th

Instructions: Solve the following 3 problems using computer software. Some notes from Dr. Hankins Computational Physics course related to air drag are provided, but if you have questions or issues, don't be afraid to ask for help. Plan to submit a short write up of questions 1 and 2 with plots and a brief description of your findings (a few sentences to a paragraph will suffice). For problem 3, I would like you to submit a more detailed write up (~3 pages in length) following a format similar to a lab report with sections including an Introduction, Methods, Results, and Discussion/Conclusions. Please add your solutions to questions 1 and 2 as an appendix to the 'main' report you are writing up for question 3. That should make it easier to submit this as a single document, and note that the appendices do not count toward the earlier specified page limits. The project assignment is worth 50 points total with 10 points each for questions 1 and 2, and 30 points for question 3. The submission link will be provided on blackboard in the 'Course Projects' folder.

1. **Problem 2.22 (Linear Air Drag):** The equation 2.39 in text (also provided below) describes the range of a projectile in a linear medium, which cannot be solved analytically (paper and pencil methods) in terms of elementary functions. If you put in numbers for the several parameters, then it can be solved numerically using any of several software packages such as Mathematica, Maple, and MatLab (I personally will suggest using Python 😊). To practice this, do the following: Consider a projectile launched at angle θ above the horizontal ground with initial speed v_0 in a linear medium. Choose units such that $v_o = 1$ and g = 1. Suppose also that the terminal speed $v_{ter} = 1$. (With $v_o = v_{ter}$, air resistance should be fairly important.) We know that in a vacuum, the maximum range occurs at $\theta = \pi/4 \sim 0.75$. (A) What is the maximum range in a vacuum? (B) Now solve (2.39) for the range in the given medium at the same angle $\theta = 0.75$. (C) Once you have your calculation working, repeat it for some selection of values of θ within which the maximum range probably lies. (You could try $\theta = 0.4, 0.5, \ldots, 0.8$.) (D) Based on these results, choose a smaller interval for θ where you're sure the maximum lies and repeat the process. Repeat it again if necessary until you know the maximum range and the corresponding angle to two significant figures (it may help to think of this as an iterative process). Compare with the vacuum values.

$$\frac{v_{yi}+v_{term}}{v_{xi}} R = v_{term}\tau \ln\left(1 - \frac{R}{v_{xi}\tau}\right) = 0 \qquad \text{(Eqn. 2.39)}$$

2. **Problem 2.44 (Quadratic Air Drag):** To get an accurate trajectory for a projectile one must often take account of several complications. For example, if a projectile goes very high then we have to allow for the reduction in air resistance as atmospheric density decreases. To illustrate this, consider an iron cannonball (diameter 15 cm, density 7.8 g/cm^3) that is fired with initial velocity 300 m/s at 50 degrees above the horizontal. The drag force is approximately quadratic, but since the drag is proportional to the atmospheric density and the density falls off exponentially with height, the drag force is $f = c(y)v^2$ where $c(y) = \gamma D^2 \exp(-y/\lambda)$ with $\gamma = 0.25$ N s^2/m^4 and λ = 10,000 m. (A) Write down the equations of motion for the cannonball and use appropriate software to solve numerically for x(t) and y(t) for $0 < t < 3.5$ s. Plot the ball's trajectory and find its horizontal range. (B) Do the same calculation ignoring the variation of atmospheric density [that is, setting $c(y) = c(0)$], and yet again ignoring air resistance entirely. Plot all three trajectories for $0 < t < 3.5$ s on the same graph. You will find that in this case air resistance makes a huge difference and that the variation of air resistance makes a small, but not negligible, difference.

3. **Launching Satellites w/ Quadratic Air Drag**: A handful of startup companies are looking to revolutionize the space industry by launching satellites from the ground with a huge initial velocity, which will take them to a specified height in the atmosphere before using any sort of rocket propulsion to carry the craft the rest of the way into orbit (see SpinLaunch as an example of this idea in action- https://www.spinlaunch.com/). This method of launching satellites effectively removes the 'first stage' of a rocket, which is a majority of the weight and fuel cost in a traditional rocket launch (think about the discussion we just had in Chp 3). If successful, this is will be a huge achievement, but many technical barriers remain before the idea can be fully realized. Imagine now for a moment, that you are part of a team working to develop a space launch system under this paradigm. You are being tasked with estimating the initial velocity of launch needed to achieve a height of 80 km above earth's surface including realistic effects such as air drag.

   Computational Methods: For this problem, I am asking you to use a different approach compared to problem #2, because that solution makes a few assumptions that we'll want to avoid for a problem of this nature. Instead, let's model the drag force in exercise using an equation known as the Prandtl expression, which is given by $F(v) = \frac{1}{2}\, c_w \rho_{air} A v^2$, where c_w is known as the drag coefficient, ρ_{air} is the density of air, A is the cross-section of the object, and v is the velocity. What makes this calculation tricky is that the density of air decreases with height in the atmosphere (as in problem 2, but you will likely want to reference a detailed model of the atmosphere such as NRLMSISE-00, which is available here: https://www.amentum.io/atmosphere_docs#tag/wam or see https://en.wikipedia.org/wiki/NRLMSISE-00) and also the drag coefficient depends on speed (and example plot is given below showing dependence of Mach number which is the ratio of speed vs. sound speed in medium). These additional dependencies on position and velocity go beyond the simple form $F(v) \propto v^2$ and make the problem much more tractable using numerical methods.

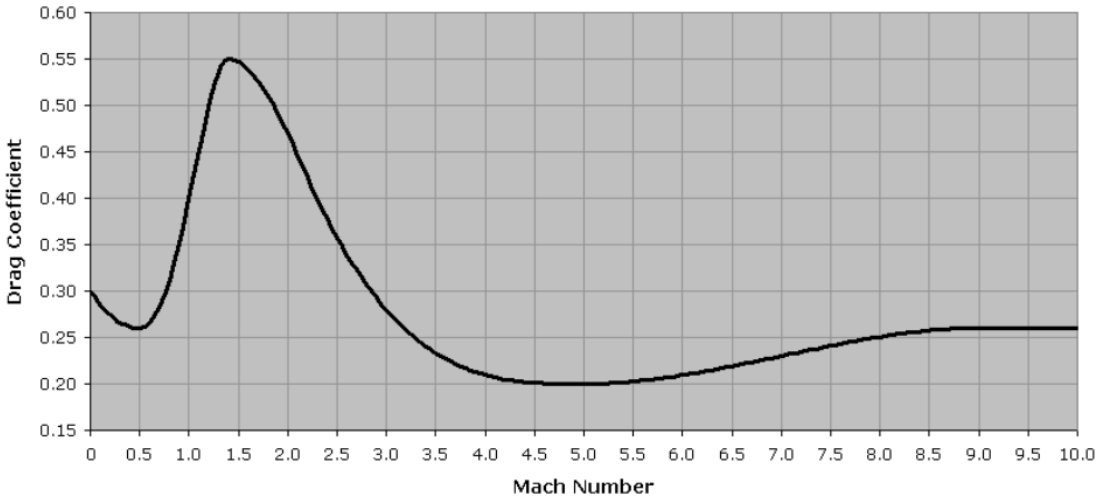
   My advice for this exercise is to think about solving the equation using a grid of points (or regions) through the atmosphere and solving the equation where we can assume simple functional forms for c_w and ρ_{air}, and then if needed 'stitching' the answers together using common boundary conditions to link one region to the next for where the functional forms of these equations might have changed from one region to the next. Note this approach will require you to think very carefully about boundary conditions and ensuring your solution doesn't violate assumptions you make about the drag coefficient and air density profile. You will also likely have to produce several trials of this calculation to determine the height achieved, and then do some iterations based on your initial velocity 'guesses' to figure out the velocity value corresponding to the desired height in the atmosphere. I'll note that I don't consider there to be a 'correct' answer to this problem, because I want to give you the freedom to think about the problem with an open mind. You are able to make choices and assumptions in this problem that make the numerics more approachable. But I will say that the better your choices and assumptions are, the more accurate your solution will be. That said, part of the grading will ask you to critically evaluate how 'reasonable' some of your choices are. If you have questions or concerns about specific items in this regard, please feel free to ask me.

   Report Grading – Your written report for this question will be graded on the following items:

   - Introduction gives sufficient background and purpose for the project
   - Methods describe the calculation including
     - Input data used
     - Assumptions/approximations employed on the data (including how reasonable they may or may not be prior to performing the calculation)
     - Numerical approach used for the calculation

- Results including plots like position, velocity vs. altitude
    - Include some written description of plots
    - Also, a good idea to revisit the idea of any assumptions post calculation to know if there could be a faulty assumption somewhere (e.g., say your projectile is at a point where the velocity change is rapid enough compared to the c_w curve that the numeric value used falls out the anticipated range)
- Discussion/Conclusions provides summary of results
    - Compare your calculation with the physics I version of this same exercise without air drag, and also think about comparing with results of a calculation repeating the methods of question #2
    - Discuss ethical considerations of this sort of launching method – can we send people, animals, biological samples, etc. to space using this sort of launch method. What are the relevant limitations in this regard?
- Additional graded items to include spelling/grammar, formatting/clarity, effective use of plots/charts.

Example Graph of Drag Coefficient



Note this particular chart was based on a calculation done for the Saturn 5 Rocket (see https://web.archive.org/web/20170313142729/http://www.braeunig.us/apollo/saturnV.htm), and the shape of the projectile does influence the calculation for c_w, but many objects show this sort of profile with peaking drag around Mach 1-2 and falling off with increasing Mach number. I am ok if you want to say that our 'thrown' object follows a very similar profile or if you want to find your own plot of c_w for a particular geometry, that's fine with me so long as you adequately describe the choice in the context provided by this exercise.